

Replication Resource Assessment and 1000-Paper Scaling Projection

REPLICATE Project

Prepared from the REPLICATE-PROJECT reports

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Abstract

This document summarizes the time, compute resources, software stack, tool challenges, and scaling implications observed in the REPLICATE-PROJECT corpus. The project contains completed and partial replications across mathematical theory, machine learning, physics simulation, computational chemistry, bioinformatics, materials modeling, astrophysics, and engineering. The tables below distinguish directly observed values from estimates. The final sections project the resources required to replicate 1000 computational papers and estimate the expansion factor required to move from replication to original scientific work.

1 Executive Summary

The completed reports show that AI-assisted replication is practical for a large fraction of computational papers, but the required resources vary by more than six orders of magnitude. Mathematical, algorithmic, and small-data machine-learning replications often complete in seconds to hours on a workstation. Simulation-heavy work, GPU deep learning, molecular dynamics, combustion, DFT, and cosmological simulations require GPU clusters, licensed software, or large HPC allocations. The main limiter is not usually writing code; it is recovering the missing operational details: exact datasets, software versions, random seeds, model hyperparameters, licensed solvers, preprocessing pipelines, and production-scale compute.

For a 1000-paper program, a realistic mixed portfolio requires a triage pipeline. A broad replication program should not assume uniform full-fidelity replication. Instead, papers should be divided into mathematical/algorithmic, workstation computational, GPU/HPC tractable, and frontier/HPC-original-scale categories. A balanced 1000-paper replication effort would likely require roughly 20–40 staff-years of technical labor, 0.5–3 million GPU-hours or equivalent for the compute-heavy subset, and a dedicated software/data operations function. Full production-level replication of all papers, including the original compute scale, could be an order of magnitude larger.

2 Scope and Method

The estimates in this document are based on local report files under REPLICATE-PROJECT, especially completed replication reports, status files, and `COMPUTATIONAL_TOOLS_INVENTORY.md`. Values labeled *observed* are reported directly in the replication documents. Values labeled *estimate* are extrapolations from observed runs, reported blockers, and common practice for the relevant scientific domain.

3 Observed Replication Time and Resource Classes

Class	Examples from project	Human time per paper	Compute per paper	Notes
Mathematical / definitional	GraphBLAS, Hausdorff integer sequences, Rice–Mele shift vector, chiral spin order	0.5–2 days <i>estimate</i>	Seconds to minutes CPU	Best fit for AI-assisted replication. Success depends on equations being complete and testable.
Small algorithmic / numerical	CPW analytical model, photo-z PDFs, FDTD reduced cases, qZSI common-mode voltage	1–3 days <i>estimate</i>	30 seconds to 1 hour CPU; occasional 0.5–2 GB RAM	Typically reproducible with Python/NumPy/SciPy. Full commercial simulations may still be out of scope.
Machine learning, reduced scale	STEM segmentation surrogate, Graph RL restoration, Bayesian optimization, latent SDE reduced model, hybrid CHF surrogate	2–7 days <i>estimate</i>	Minutes to hours CPU/GPU	Often requires surrogate data because original training data are missing or too large. Exact reproduction is limited by data and hyperparameter ambiguity.
GPU deep learning / generative AI	PVMol-Gen, neural VMC, chaotic neural ODE	1–3 weeks <i>estimate</i>	Minutes to tens of GPU-hours for reduced runs; tens to hundreds for full runs	Environment setup, model instability, and data reconstruction dominate effort.
Computational chemistry / materials	Pt/La ₂ Ti ₂ O ₇ DFT, 2D GaN, perovskite molecule discovery	1–4 weeks <i>estimate</i>	Tens to thousands of GPU/CPU-hours	Exact replication depends on pseudopotentials, exchange-correlation functional, k-mesh, relaxation criteria, slab models, and licensed or specialized codes.
Bioinformatics pipelines	Virophage taxonomy, metagenome binning, protein clustering	3 days–3 weeks <i>estimate</i>	Hours to days CPU; 10s–100s GB storage for full datasets	Toolchains are broad and brittle. Reference databases, version drift, and network/database access are the main risks.
Molecular dynamics / Markov models	MSM/OOM alanine dipeptide, VMC, protein/trajectory analyses	1–4 weeks <i>estimate</i>	Observed MSM: about 2 h short trajectories + 4 h long reference on 8 A100s; analysis <5 min	Reproducibility depends on force fields, initial states, trajectory length, thermostat/barostat, and random seeds.

Class	Examples from project	Human time per paper	Compute per paper	Notes
Large-scale PDE / combustion / cosmology	Ignition kernel, cosmic reionization, full non-Markovian FDTD, production turbulent flow	1–3 months <i>estimate</i>	10^4 – 10^6 hours or 10s–1000s GPU-hours	Simplified replication is possible; full-fidelity replication requires production HPC workflows.

Table 1: Observed and estimated resource classes for papers in the REPLICATE corpus.

4 Representative Report-Level Resource Notes

Replication report	Status	Observed compute	com-	Observed / implied human effort	Primary limitation
GraphBLAS mathematical foundations	Fully confirmed	Test suite <1 s CPU		1–2 days <i>estimate</i>	Mainly implementation discipline; paper is definitional and internally testable.
Photo-z PDF compression	Confirmed	54 s CPU <i>observed</i>		1–2 days <i>estimate</i>	Original BPZ mock catalogs unavailable; synthetic GMM catalogs used.
CPW resonator mask	Substantially confirmed	27 s total CPU; sparse solve 2.5 s <i>observed</i>		1–3 days <i>estimate</i>	Full 3D FEM / HFSS-style analysis not reproduced.
FDTD delay PDE	Substantially confirmed	266 s + 52 s + 44 s CPU for reported reduced cases <i>observed</i>		2–4 days <i>estimate</i>	Full non-Markovian case estimated at 50–125 GB memory.
Bayesian optimization qHSRI	Partially confirmed	About 42 min CPU <i>observed</i>		3–5 days <i>estimate</i>	Different implementation stack; omitted noisy/multi-objective/application cases.
Chiral spin order	Strong qualitative confirmation	<10 s CPU <i>observed</i>		1–2 days <i>estimate</i>	Mean-field theory replicated; beyond-mean-field critical exponent not captured.
Trapped-ion light scattering	Core analytic result confirmed	Small ODE tests CPU; all tests pass <i>observed</i>		3–5 days <i>estimate</i>	Quantitative figures require Penning-trap mode simulation details not fully specified.
STEM deep learning segmentation	Substantial reduced replication	95 s on one A100 80 GB <i>observed</i>		3–7 days <i>estimate</i>	Synthetic images replace experimental/multislice image stacks.
PVMol-Gen	Partially confirmed	8 A100s for SELFIES pipeline; generation about 8.5 h for a major stage <i>observed</i>		1–3 weeks <i>estimate</i>	Stage 1 classifier below paper metric; DFT/xTB filters and SMILES-X details are major uncertainties.

Replication report	Status	Observed compute	com-	Observed / im- plied human effort	Primary limitation
Neural VMC for light nuclei	Partial method confirmation	Single A100 40 GB; total wall time <150 s for reported runs <i>observed</i>		3–7 days <i>estimate</i>	Simpler Minnesota interaction used instead of full AV6’/AV8’ operator structure.
MSM/OOM bias correction	Strong confirmation	8 A100 80 GB; short ensemble about 2 h, long reference about 4 h, analysis <5 min <i>observed</i>		1–2 weeks <i>estimate</i>	Exact reference trajectories and MD settings drive reproducibility.
Latent SDE quasar variability	Reduced method replication	160 s CPU vs paper about 6 weeks on V100 <i>observed</i>		3–7 days <i>estimate</i>	Full six-band LSST-like data generation and GPU-scale training not reproduced.
Hybrid CHF ML	Partial confirmation	About 7 min on 8-core CPU <i>observed</i>		3–5 days <i>estimate</i>	NRC CHF database not redistributable; synthetic surrogate used.
Cosmic reionization on computers	Semi-analytic replication	About 3 min CPU, <2 GB RAM <i>observed</i>		3–7 days <i>estimate</i>	Original CROC simulations require about 10^4 cores and months; surrogate FGPA model used.
Ignition kernel turbulent combustion	Simplified attempt failed; GPU framework partial	Original about 40,000 CPU-h per case, 20 cases about 800,000 CPU-h <i>observed</i> ; GPU estimate about 80 A100 GPU-h for full open-source run <i>estimate</i>		1–3 months <i>estimate</i>	Requires compressible reacting LES with detailed chemistry; low-Mach solver was physically mismatched.

Table 2: Representative resource notes extracted from the reports.

5 Comprehensive Tool and Software Inventory

Category	Tools	Replication role	Challenges
Core scientific Python	Python, NumPy, SciPy, pandas, matplotlib, h5py, xarray, netCDF tools	Most numerical analysis, plotting, data processing, reduced physics models	Version drift, binary wheels on unusual architectures, silent numerical differences.
Testing / workflow	pytest, shell scripts, Make, LaTeX, Jupyter notebooks	Regression tests, figure generation, reproducible reports	Notebooks often lack deterministic execution; scripts need dependency pinning.
Machine learning	PyTorch, TensorFlow, Keras, scikit-learn, GPyTorch, BoTorch, TensorFlow Probability, Ray Tune	Deep learning, Bayesian optimization, uncertainty models, neural ODE/SDE models	GPU/CUDA compatibility, nondeterminism, hyperparameter ambiguity, deprecated APIs.
Generative AI / language models	Hugging Face Transformers, GPT-2, LLaMA-2, RoBERTa/SciBERT, BERTopic, HDBSCAN, UMAP	Molecular generation, domain language models, embeddings and clustering	Model weights, tokenizer variants, GPU memory, licensing, exact pretraining corpora unavailable.
Cheminformatics	RDKit, SMILES-X, SELFIES, PubChem API, PAINS/SA filters, Morgan fingerprints	Molecular classification, generation, filtering, clustering	SMILES-X reproducibility gap, API dependence, molecular standardization differences, missing DFT/xTB filters.
DFT / electronic structure	Quantum ESPRESSO, VASP, PBE/PBE0, PAW/SSSP pseudopotentials, Bader analysis, VESTA	Materials band structures, DOS/PDOS, adsorption and slab calculations	VASP license; QE GPU build quirks; pseudopotential and smearing choices change results.
Molecular dynamics	OpenMM, AMBER ff14SB/TIP3P, ACEMD, LAMMPS, EAM potentials, OVITO/Voronoi analysis	Trajectory generation, Markov state models, materials deformation	Long trajectories, force-field versioning, initial conditions, thermostat/barostat choices.
Combustion / fluid dynamics	PeleC, PeleLMeX, AMReX, SUNDIALS/CVODE, DRM19 chemistry, CharLES ^X reference	Reacting-flow LES and ignition-kernel dynamics	Proprietary original solver; low-Mach vs compressible physics mismatch; stiff chemistry and GPU runtime failures.

Category	Tools	Replication role	Challenges
Nuclear / reactor physics	SCALE 6.3, ORIGEN, TRITON, KENO-VI, NEWT, Serpent, ENDF/B data	Depletion and neutron transport studies	Licenses/distribution controls, nuclear data library versions, high compute cost.
Electromagnetic / circuit simulation	Ansys HFSS, COMSOL, Palace, MATLAB/Simulink, PySpice/ngspice	CPW resonators, qZSI common-mode voltage, FEM/eigenmode simulations	Commercial licenses; geometry/mesh details underspecified; open-source approximations often lower fidelity.
Bioinformatics	HMMER/hmmsearch, BLAST+, DIAMOND, MAFFT, MUSCLE, IQ-TREE, clipkit, Prodigal, MCL, InfoMap, MUMmer, CheckV, metaSPAdes, MetaBAT2, minimap2, samtools	Taxonomy, genome binning, alignments, phylogenies, metagenomic assembly	Large databases, version sensitivity, long install chains, network/download brittleness.
R ecosystem	R, hetGP, DiceOptim, GPareto, DiceKriging, mlrMBO, mco, pso, fldgen, ggplot2, dplyr, tidyr, ggtree	Bayesian optimization, climate emulation, plotting, phylogenetics	Some packages are old or R-only; reproducing R results in Python can alter rankings.
Parallel / HPC	MPI, OpenMP, POSIX threads, CUDA, OpenCL, multiprocessing, SLURM/PBS	Scaling experiments, GPU training, trajectory generation, CFD/DFT/MD	Queue delays, module incompatibilities, GPU driver failures, hardware-specific performance.
Databases / external data	OEIS, NCBI GenBank/RefSeq/nr, IMG/VR, Pfam, PDB70, PubChem, IRPhEP, ENDF/B, experimental proprietary databases	Inputs, benchmarks, labels, reference datasets	Access restrictions, nonredistributable data, changing database snapshots.

Table 3: Software and tool inventory with recurring replication challenges.

6 Common Replication Challenges

Challenge	Observed impact	Mitigation for a scaled program
Missing datasets	Several reports used synthetic surrogates because original catalogs, experimental databases, or simulation outputs were unavailable.	Data acquisition team; DOI/dataset tracking; institutional agreements; snapshot mirrors.
Licensed / proprietary software	VASP, SCALE, MATLAB/Simulink, HFSS/COMSOL, CharLES ^X block exact reproduction.	Maintain licensed software pool; define acceptable open-source substitute criteria; record fidelity level.
Underspecified parameters	Pseudopotentials, random seeds, trap modes, mesh settings, optimizer settings, and filtering thresholds often missing.	Standard replication metadata checklist; structured uncertainty log; sensitivity analysis.
Version drift	Old Python/R APIs, GPU builds, CUDA compatibility, pandas changes, and binary package issues caused failures.	Container registry; lockfiles; reproducible build recipes; long-term artifact archive.
Compute mismatch	Full-fidelity combustion, cosmology, DFT, and MD can be orders of magnitude larger than reduced replication.	Tiered replication: reduced, comparable, and production-fidelity levels.
Benchmark ambiguity	Some papers report figures without exact numeric data or use private baselines.	Digitization plus uncertainty bands; ask authors when possible; report qualitative/quantitative status separately.
Non-determinism	ML and stochastic simulations vary with seed, hardware, libraries, and thread count.	Multi-seed protocols; confidence intervals; deterministic flags where possible.
Toolchain breadth	Bioinformatics and HPC workflows require many tools, databases, and build systems.	Domain-specific environment templates; staff specialists by domain; automated smoke tests.

Table 4: Recurring replication challenges and mitigation strategies.

7 Projection for Replicating 1000 Papers

7.1 Portfolio Assumption

A plausible 1000-paper replication portfolio will be heterogeneous. The following projection assumes a mix similar to the REPLICATE corpus after triage toward computationally tractable papers.

Paper class	Share	Count	Staff-days / paper	Compute / paper
Mathematical / algorithmic	25%	250	1–2	<1 CPU-hour

Paper class	Share	Count	Staff-days / paper	Compute / paper
Workstation numerical / small ML	35%	350	2–5	1–100 CPU-hours
GPU ML / medium simulation	20%	200	5–15	1–200 GPU-hours
Domain pipelines / databases	10%	100	7–20	100–5,000 CPU-hours plus storage
Heavy HPC / full simulation	10%	100	15–60	10^4 – 10^6 CPU-hours or 100–5,000 GPU-hours

Table 5: Assumed 1000-paper mix after computational-replicability triage.

7.2 Aggregate Resource Estimate

Program level	Human labor	Compute	Interpretation
Screening only	1–3 staff-years	Small workstation cluster	Identify replicability, software stack, data availability, and likely cost for 1000 papers.
Reduced replication	20–40 staff-years	0.1–0.5 million GPU-hour equivalent; 1–10 million CPU-hours	Rebuild core claims with reduced data, surrogate datasets, and lower-fidelity simulations where needed.
Comparable-fidelity replication	50–100 staff-years	0.5–3 million GPU-hour equivalent; 10–100 million CPU-hours	Aim to reproduce most figures/tables at comparable scale where data/software are accessible.
Production-fidelity replication	100–250 staff-years	3–20 million GPU-hour equivalent; 0.1–1 billion CPU-hours	Full original-scale reproduction, including large PDE, MD, DFT, cosmology, and combustion runs.

Table 6: Aggregate 1000-paper replication program estimate. One staff-year is approximated as 220 working days.

7.3 Staffing Model

Role	Headcount for 1000-paper reduced/comparable program	Responsibilities
Replication scientists / research software engineers	12–25	Paper triage, implementation, tests, figure generation, documentation.
Domain specialists	5–12	DFT, bioinformatics, ML, physics simulation, MD, nuclear, combustion, astrophysics review.
HPC / DevOps engineers	3–6	Cluster jobs, containers, CUDA/MPI builds, storage, CI, artifact archiving.
Data acquisition / curation	2–5	Dataset recovery, database snapshots, licensing, metadata.
Technical editors / QA reviewers	2–4	Report normalization, evidence grading, audit trail, reproducibility checklists.
Program lead / project management	1–3	Prioritization, scheduling, risk management, author outreach.

Table 7: Indicative staffing for a 2–3 year 1000-paper effort.

8 Expansion from Replication to Original Work

Original work is not just “more replication.” It adds hypothesis generation, experimental design, parameter exploration, validation against new data, uncertainty quantification, writing, and peer-review response. The expansion factor depends strongly on domain.

Domain type	/ work	Replication baseline	Original-work multiplier	Why original work expands
Mathematical	/ algo-rithmic	1–2 days	3–10×	Need new theorem/algorithm, proof search, counterexamples, broader test cases, and exposition.
Small method	numerical	2–5 days	5–15×	Need parameter sweeps, ablations, robustness tests, and comparison against alternatives.
Machine learning		1–2 weeks	5–20×	Need dataset construction, model selection, tuning, multi-seed evaluation, baselines, and generalization tests.
DFT/materials		1–4 weeks	10–50×	Need structure search, multiple surfaces/defects, convergence studies, adsorption pathways, higher-level validation.
MD / molecular simulation		1–4 weeks	10–50×	Need longer sampling, force-field validation, multiple systems, uncertainty quantification.

Domain / work type	Replication baseline	Original-work multiplier	Why original work expands
Bioinformatics	1–3 weeks	5–25×	Need new database curation, biological validation, phylogenetic sensitivity, manual annotation.
Combustion / large PDE / climate / cosmology	1–3 months	10–100×	Need production simulations, ensemble runs, mesh/time-step convergence, validation data, and large-scale analysis.

Table 8: Estimated expansion factor from replication to original work.

8.1 1000-Paper Equivalent Original-Work Projection

If the same topical distribution were used not for replication but for producing original publishable contributions, the total effort would likely expand by a factor of 8–30 for a conservative program and 30–100 for frontier simulation-heavy work. A 1000-paper reduced/comparable replication effort estimated at 20–100 staff-years therefore maps roughly to:

Original-work ambition	Staff-years	Compute multiplier	multi-	Likely output
Incremental extensions	150–400	3–10×		Add ablations, variants, new baselines, modest parameter sweeps, or additional datasets.
New publishable studies	300–1500	10–50×		Produce robust new results in selected domains, with peer-review-grade validation.
Frontier original science at scale	1000+	50–200×	for heavy subsets	New large simulations, new datasets, experimental campaigns, or production HPC campaigns.

Table 9: Approximate expansion from a 1000-paper replication program to original research programs.

9 Operational Recommendations

1. Use a three-tier evidence label for every paper: **reduced replication**, **comparable-fidelity replication**, and **production-fidelity replication**. This prevents a fast surrogate from being mistaken for exact reproduction.
2. Build a reusable software environment library organized by domain: Python ML, R optimization, bioinformatics, DFT, MD, combustion, nuclear, and LaTeX/reporting.

3. Require every replication to produce: executable scripts, dependency lockfile or container recipe, output checksums, key figures/tables, test suite when possible, and a structured limitations table.
4. Maintain a central data registry with dataset status: public, mirrored, author-provided, licensed, nonredistributable, synthetic substitute, or missing.
5. Reserve HPC for papers that pass a low-cost pilot. Many core claims can be tested with a reduced model before committing thousands of GPU-hours.
6. Separate engineering success from scientific success. A software pipeline may run correctly while failing to reproduce the original claim because of missing data, inappropriate surrogate physics, or underspecified parameters.

10 Bottom Line

The REPLICATE corpus supports a practical conclusion: AI-assisted replication can make 1000-paper computational replication plausible, but only with triage, evidence grading, domain specialists, and serious environment/data management. The median paper is manageable on workstation or modest GPU resources. The tail dominates cost: a small number of combustion, cosmology, DFT, MD, and large-model papers can consume most compute and staff time. Original work expands the resource envelope substantially, usually by one order of magnitude and sometimes by two, because it requires exploration rather than reconstruction.